

RoboticsNAS: Closed-Loop Architecture Search for VLA Models

Status summary, 2026-10-10 | SmolVLA on LIBERO (4 suites, 40 tasks) | code and logs:
github.com/Autzoko/RoboticsNAS

Bottom line. Offline validation loss does not predict closed-loop success of VLA architectures, while cheap closed-loop-aware signals do. Tuning the inference schedule of the unchanged SmolVLA network gives +14.7 points on the official LIBERO test. Several much smaller networks match or beat the default when trained standalone, but weight-sharing supernets cannot identify them (4 fixes tried, all fail). **Adopted model:** SmolVLA default network with 2 denoising steps and 10 executed actions per call (official test 0.821 vs 0.674 published). **Strongest candidate (not yet officially tested):** v12-e4-stretch-f1-t16 (search-val 0.880, 5.7x faster per call, 37% less deploy memory).

1. Setup and protocol

- Base: *lerobot/smolvla_base* (no LIBERO pretraining); VLM frozen, action expert trained. Elastic search space of 2,448 configs: VLM depth, expert depth/width, VLM-to-expert bridge, visual tokens, denoising steps, executed horizon. Key format v{VLM layers}-e{expert layers}-{bridge}-f{FFN}-t{tokens}-s{steps}-h{horizon}.
- Leakage control: train-only normalization stats; search on freshly sampled scenes with dedicated seeds; official init states touched once, for candidates frozen in git beforehand.

2. Offline proxies vs closed-loop success (111 archs x 400 episodes)

Ranking signal	Kendall tau vs held-out SR	Top-10 overlap
Split-half ceiling (closed loop, 200 vs 200 eps)	0.840	8/10
Closed loop, 2 episodes / task	0.819	6/10
Agreement with anchor policy on visited states (ours, 0 rollouts)	0.738	8/10
NASWOT (zero-cost)	0.523	-
SNIP (zero-cost)	0.322	-
Flow-matching validation loss (standard NAS signal)	0.165	0/10
Grad-norm (zero-cost)	-0.51 (inverted)	-

Across training of one network, validation loss rises 0.515 to 0.961 while closed-loop success rises 0.70 to 0.85 (10k to 30k steps): selecting by loss would pick the worst checkpoint.

3. Search under a rollout budget (replay, 300 repetitions)

Method (regret vs best eligible arch, 400 eps)	No cap	ms/step <= median
Proxy shortlist + paired successive halving (ours)	0.058	0.024
NASWOT shortlist + paired successive halving	0.034	0.024
Successive halving, random pool	0.089	0.080
Learned predictor (ridge, BO-style)	0.092	0.098
Validation-loss shortlist	0.130	0.112
Random search	0.181	0.178

A bound-based racing variant (bootstrapped reference, M1+M3) gave no consistent gain over the simple shortlist + racing; it under-explores.

4. Official LIBERO test (2,000 episodes each, frozen candidates)

Configuration	Network	Schedule	Success	ms/call	ms/step
SmolVLA as published	v16-e16-stretch-f1-t64	10 steps, exec 50	0.674	396	7.9
Adopted: default net, tuned schedule	v16-e16-stretch-f1-t64	2 steps, exec 10	0.821	127	12.7
Same schedule, standalone weights	v16-e16-stretch-f1-t64	2 steps, exec 10	0.810	127	12.7
Small standalone net	v12-e4-stretch-f1-t16	4 steps, exec 5	0.823	88	17.7
Standard NAS (val-loss pick)	v16-e8-stretch-f1-t16	4 steps, exec 50	0.701	130	2.6
Random search pick	v16-e16-stretch-f1-t64	10 steps, exec 50	0.662	396	7.9

Latency: A100-40GB, batch 1, eager bf16.

5. Standalone ground truth and efficiency (16 nets, 2 steps / exec 10, search-val)

Network	Success	ms/call	ms/step	Active params	Deploy memory
SmolVLA published (v16-e16, 10 steps, exec 50)	0.713	396	7.9	403M	980 MB

Network	Success	ms/call	ms/step	Active params	Deploy memory
Default net v16-e16, tuned schedule	0.843	127	12.7	403M	980 MB
v12-e4-stretch-f1-t16	0.880	69	6.9	290M	616 MB
v8-e16-stretch-f0.75-t16	0.880	113	11.3	306M	654 MB
v16-e12-top-f0.5-t16	0.875	112	11.2	352M	726 MB
Other 11 nets	0.752 - 0.863	-	-	-	-

Most networks are within ~ 0.05 success of each other; the meaningful objective is cost at statistically tied success.

6. Weight-sharing supernet (Gate 1): robust negative result

Supernet variant	Kendall vs standalone (n=16)	Mean SR gap	Gap, 4-8 layer experts
V0 original (sandwich + distillation)	0.077	+0.198	+0.254
V1 per-depth norm gains	0.051	+0.194	+0.245
V3 PCGrad (gradient-conflict projection)	0.068	+0.190	+0.248
V4 bridge-conditioned K/V adapters	-0.034	+0.182	+0.247

- Noise ceiling: test-retest Kendall of the standalone ranking itself is ~ 0.56 ; supernets recover $\sim 0-25\%$ of it.
- Diagnosis: gradient conflict sits in the shared expert trunk (cosine 0.17 between 4- and 16-layer subnets); the readout is not the cause (readout-only fine-tune: no gain); the gap is explained by bridge mismatch (R^2 0.52) more than by expert depth (0.35). A bridge-indexing bug was found and fixed.
- Example: the best standalone net v12-e4-stretch-f1-t16 (0.880) is scored only 0.44-0.54 by the four supernets.

7. What the results support, and next step

- **Supported now (analysis / benchmark paper):** offline loss fails for VLA architecture evaluation; closed-loop-aware proxies and paired racing work; inference schedule is a first-class axis; latency and memory measured; weight-sharing failure rigorously diagnosed.
- **Not yet supported:** a NAS method that discovers better architectures by itself.
- **Proposed next step (awaiting decision):** two-stage method - supernet + proxy shortlist cost-diverse candidates, short standalone training ($\sim 10k$ steps), paired closed-loop race for the cheapest net tied with the best, frozen official test. First check reuses 13 existing 10k-step checkpoints (evaluation only). Alternatives: run the last supernet variant V2 (~ 8 GPU-h) or stop and write up.